

Coexistence Downlink Interference Analysis Between LEO System and GEO System in Ka Band

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Abstract—With the development of satellite communication techniques, various types of new service have been emerging, which leads to more spectrum resource requirements. However, traditional L(1-2GHz)/S(2-4GHz) bands which are allocated to satellite communications are limited, and these bands cannot meet the increasing bandwidth demands. Therefore, the Ka(27-40GHz) band has attracted the attention of many satellite communication institutes due to the large bandwidth. With the implementation of many Geostationary Earth Orbit (GEO) and Low Earth Orbit (LEO) constellation plans in Ka band, the coexistence interference analysis between LEO and GEO system is important. In this paper, the coexistence downlink interference between LEO system and GEO system is analyzed, and the impact of the exclusive angel strategy on the inference is evaluated. Simulation results show that the exclusive angel strategy can reduce the in-line downlink interference from LEO to GEO. However, it decreases the coverage of LEO. In addition, Simulation results show that there is little downlink interference from the GEO to LEO.

Keywords—downlink interference, satellite communication, exclusive angel, Ka band.

I. INTRODUCTION

Satellite communication system is an important part of modern communication. Compared with other communication technologies, satellite communication has the advantages of long-distance communication, wide coverage area, relay transmission, multiplexing technology, and stable performance for transmission. It is able to communicate without considering terrain and distance. As a result of which, satellite communication systems are widely used worldwide^[1].

In satellite communication systems, the spectrum resources are the basis for stable operation of satellite systems and it affects both the system capacity and the communication rate. However, the spectrum resources of satellite communications are limited. The traditional L(1-2GHz)/S(2-4GHz)/Ku(12-18GHz) bands for satellite mobile communications are almost occupied fully. Therefore, the Ka(27-40GHz) band is used to obtain the large bandwidth for both Geostationary Earth Orbit (GEO) and Low Earth Orbit (LEO) satellite systems. For example, the business companies such as OneWeb and O3b have already proposed their own LEO constellation plan. However, the large-scale deployment of satellite systems (e.g.

720 LEO satellites in OneWeb constellation) will cause co-frequency interference between LEO system and GEO system. Therefore, it is of great importance to analyze the co-frequency interference between LEO and GEO systems.

In the co-frequency downlink interference from LEO system to GEO system, the most serious interference is in-line interference. There are many researches objecting at the in-line issue in the published paper. The interference analysis between GEO and MEO satellite systems is presented in [2], and it proposes an adaptive power control technique in order to mitigate the in-line interference. A radio frequency interference analysis based on simulation of the dynamic satellite constellation and the designed parameters of the inter-satellite links are analyzed in [3]. Range-based, traffic-aware, and cognitive power control techniques are presented in [4]. Considering the spectral co-existence of GEO and Non-Geostationary Earth Orbit (NGEO), a cognitive range-based power control algorithm is proposed in [5], and it shows that the proposed cognitive power control algorithm can mitigate the harmful in-line interference on the GEO terminal receiver, while also providing the desired link quality for the NGEO system. It is noteworthy that these papers mentioned above mainly analyze the effect of power control strategies on the suppression of coexistence interference. However, the exclusive angel strategy presented in [6] is valuable for removing in-line interference and it should be taken into consideration in the analysis.

This paper concentrates on analyzing the downlink interference between LEO system and GEO system in Ka band. The main contributions of this paper are:

- (1) The downlink interference scenarios are introduced, and the case of maximum interference called ‘worst case’ is presented.
- (2) The effect of exclusive angle strategy on the in-line downlink interference and the coverage of LEO system is analyzed.
- (3) The simulation results of the downlink interference between LEO and GEO systems are obtained under the condition of using OneWeb constellation.

The paper is structured as follows. Section II introduces the system model of GEO and LEO. Section III analyzes the

coexistence interference. Section IV describes the simulation parameter and presents the simulation results. Section V draws the conclusion.

II. SYSTEM MODEL

A. GEO System

The GEO system consists of two parts: the GEO satellite and the GEO earth station. Based on the geocentric coordinate system, we define the number of GEO to be 1, referring to [7], and its coordinate is expressed as

$$\begin{bmatrix} x_{GEO} \\ y_{GEO} \\ z_{GEO} \end{bmatrix} = (Re + h_{GEO}) \begin{bmatrix} \cos(Long_{GEO}) \\ \sin(Long_{GEO}) \\ 0 \end{bmatrix}, \quad (1)$$

where Re is the radius of the earth, and h_{GEO} is the altitude of the GEO. $Long_{GEO}$ is the longitude under the GEO satellite. The GEO transmit antenna pattern is referred to [8].

GEO earth station is the terrestrial gateway station communicating with GEO. We define the number of GEO earth stations as 1 and its coordinate is expressed as

$$\begin{bmatrix} x_{GES} \\ y_{GES} \\ z_{GES} \end{bmatrix} = (Re + h_{GES}) \begin{bmatrix} \cos(Lat_{GES}) \cos(Long_{GES}) \\ \cos(Lat_{GES}) \sin(Long_{GES}) \\ \sin(Lat_{GES}) \end{bmatrix}, \quad (2)$$

where h_{GES} is the height of the GEO earth station from the ground, Lat_{GES} is the latitude of the GEO earth station, and $Long_{GES}$ is the longitude of the GEO earth station. In downlink interference analysis, the GEO earth station receive antenna pattern is referred to [9].

The communication strategy of the GEO system simulation model is single-transmit and single-receive. The coverage of the GEO satellites can be determined from the minimum elevation angle. This article will establish a GEO earth station for the 'worst-case' scenario. The detailed analysis of the 'worst case' scenario is presented in Section III.

B. LEO System

The LEO system includes LEO constellation and LEO earth station. The LEO constellation has the characteristics of short transmission delay, low path loss, global coverage, etc. Constellation simulation depends on the orbit parameters (Semi-major axis r / Inclination I / Right ascension of the ascending node Ω / Argument of perigee ω / True anomaly E / Eccentricity e). In the downlink interference analysis, the transmit antenna pattern of the LEO satellite refers to [10]. Coordinate of LEO is expressed as

$$\begin{bmatrix} x_{LEO} \\ y_{LEO} \\ z_{LEO} \end{bmatrix} = \begin{bmatrix} r[\cos(\Omega) \cos(E) - \sin(\Omega) \cos(I) \sin(E)] \\ r[\sin(\Omega) \cos(E) + \cos(\Omega) \cos(I) \sin(E)] \\ r[\sin(I) \sin(E)] \end{bmatrix}. \quad (3)$$

The number of LEO earth stations is set to be 1, and its coordinate is calculated in the same way as GEO earth station.

The downlink communication strategy of LEO system model is defined to be that the LEO earth station selects the LEO with the highest elevation angle at the moment, communicating with the LEO by point to point.

III. INTERFERENCE ANALYSIS

In this paper, considering the co-frequency interference in Ka-band, two types of downlink interference conditions are analyzed: downlink interference from LEO to GEO, downlink interference from GEO to LEO.

A. Downlink interference from LEO to GEO

According to [6], exclusion angle strategy can be used to avoid the in-line downlink interference from LEO to GEO. As shown in Fig.1, the exclusion angle is defined as the angle between the line from LEO to GEO earth station and the GEO main antenna axis. The area which is the angle α off the GEO main antenna axis is called exclusive zone. Exclusive angle can be calculated by

$$\alpha = \arccos \frac{r_1 \cdot r_2}{|r_2| |r_1|}. \quad (4)$$

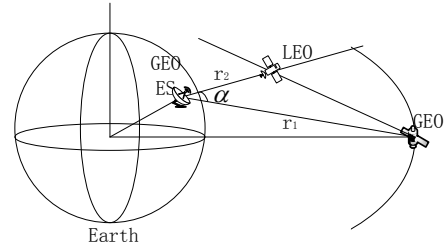


Fig. 1. Define the exclusive angel

The exclusive zone strategy means that if the LEO is inside the exclusion zone, it is not allowable to transmit signal., and if the LEO is outside the exclusion zone, it is allowable to communicate.

Using the exclusive angel strategy, the downlink interference from the LEO to the GEO is shown in Fig.2.

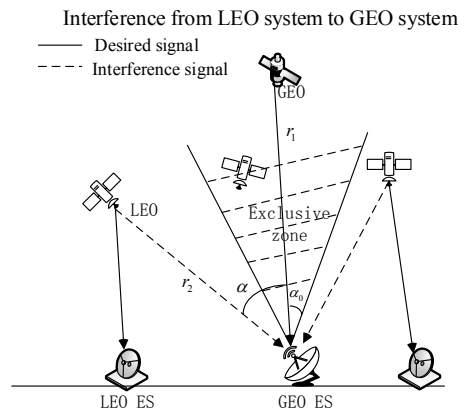


Fig. 2. Downlink interference from LEO to GEO

We analyze the scenario that caused the maximum interference, called the ‘worst case’. It is divided into three aspects

- (1) The GEO earth station is established at the position of the last beam antenna spindle of the GEO satellite coverage edge.
- (2) GEO earth station-LEO satellite-GEO satellite in-line.
- (3) The GEO earth station is co-located with the LEO earth station.

We simulated the ‘worst case’ scenario which is shown in Fig.3.

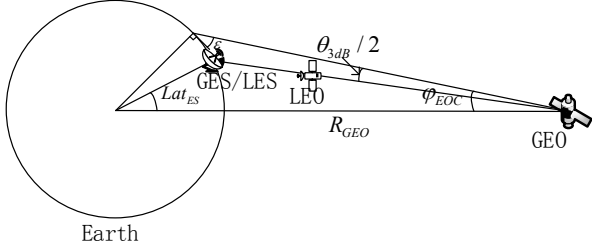


Fig. 3. The worst case (LEO interferes GEO earth station)

The advantage of the exclusive zone strategy is that it can remove the in-line downlink interference. However, the setting of the exclusion zone decreases the coverage of LEO, and it forms a non-communication zone which changes with the latitude of the GEO earth station. It is shown in Fig.4.

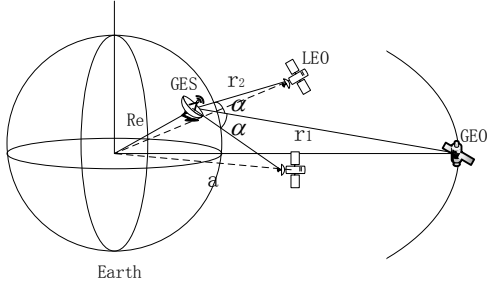


Fig. 4. LEO latitude range in exclusive zone

The maximum LEO latitude at the edge of the exclusion zone is calculated as

$$\beta_1 = \arcsin(R_e / a \times \cos(el + \alpha)), \quad (5)$$

$$Lat_{LEO_{max}} = Lat_{GES} - (90 - \alpha - el - \beta_1). \quad (6)$$

The minimum LEO latitude at the edge of the exclusion zone is calculated as

$$\beta_2 = \arcsin(R_e / a \times \cos(el - \alpha)), \quad (7)$$

$$Lat_{LEO_{min}} = Lat_{GES} - (90 + \alpha - el - \beta_2). \quad (8)$$

Therefore, LEO satellites is not allowable to communicate in the range from $Lat_{LEO_{min}}$ to $Lat_{LEO_{max}}$.

According to the provisions of [9], it suggests to evaluate downlink interference to GEO systems by $EPFD$. The formula of $EPFD$ is

$$EPFD = 10 \log_{10} \left[\sum_{i=1}^N 10^{10} \frac{P_i}{4\pi d_i^2} \frac{G_t(\theta_i)}{G_{r_{max}}} \right], \quad (9)$$

where P_i is the downlink interference transmit power, d_i is the distance from the interferer to the interferer, G_t is the transmit antenna gain, G_r is the receive antenna gain, θ_i and φ_i are the off-axis angles of the transmit and receive antennas respectively, and $G_{r_{max}}$ is the peak gain of the receive antenna.

B. Downlink interference from GEO to LEO

The downlink interference from GEO system to LEO system is shown in Fig. 5.

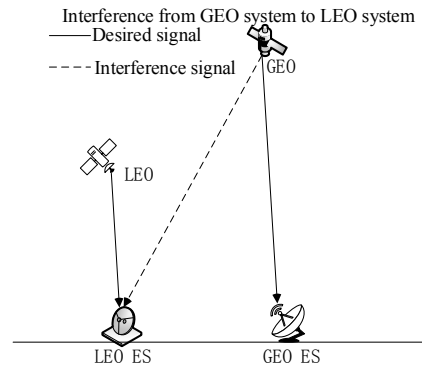


Fig. 5. Downlink interference from GEO to LEO

The ‘worst case’ scenario is defined to be

- (1) The GEO earth station is established at the GEO sub-satellite point.
- (2) The GEO earth station is in the same location as the LEO earth station.
- (3) The LEO earth station receiving antenna points to the LEO satellite with the lowest visible elevation angle.

We set the ‘worst case’ scenario as it shown in Fig 6.

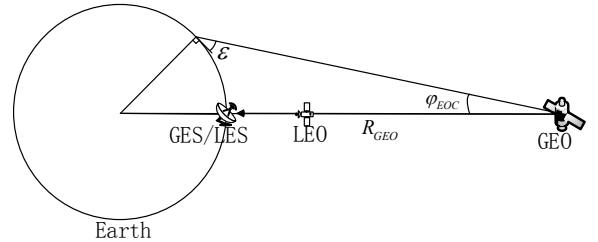


Fig. 6. The worst case (GEO interferes LEO earth station)

According to [11], the downlink interference from the GEO system to the LEO system can be expressed by the value of C/I, it is calculated as

$$(C/I)_{down} = 10 \log_{10} \left(\frac{P_{LEO} \times G_{LEO}(\theta_1) \times G_{LEOES}(\theta_2) \times I_{LEO}}{P_{GEO} \times G_{GEO}(\theta_3) \times G_{GEOES}(\theta_4) \times I_{GEO}} \right), \quad (10)$$

where P_{LEO} is the LEO transmit power, P_{GEO} is the GEO transmit power, G_{LEO} and G_{GEO} are the LEO and GEO satellite transmit antenna gains, G_{LEOES} and G_{GEOES} are the receive antenna gains of the LEO earth station and the GEO earth station, $\theta_1/\theta_2/\theta_3/\theta_4$ are the off-axis angles of the correspond antennas, I_{LEO} is the transmission loss of the LEO carrier signal, and I_{GEO} is the transmission loss of the GEO interference.

C. Simulation process

In order to calculate the co-existence interference between LEO and GEO system, as shown in Fig.7, the flow of simulation calculation is given.

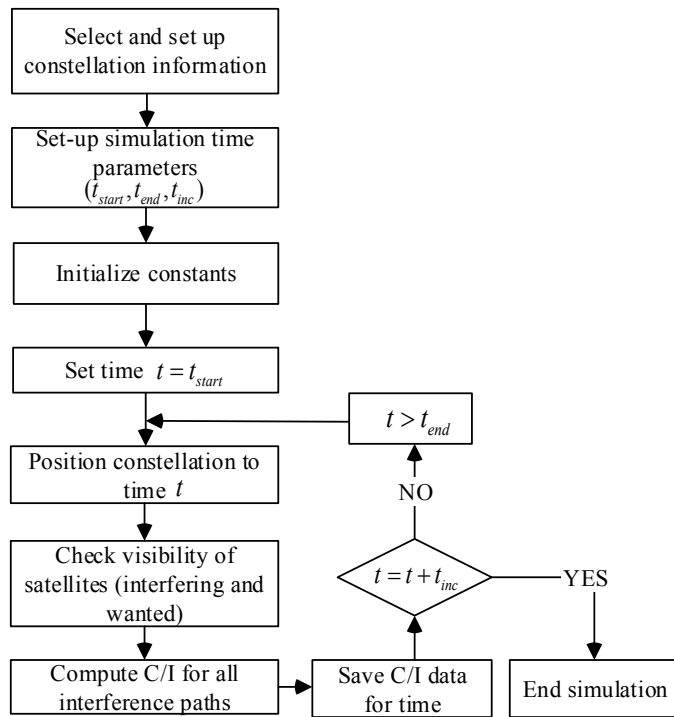


Fig. 7. The flow of simulation calculation

IV. SIMULATION RESULTS

A. LEO and GEO Systems Parameters

The parameters of GEO system are shown in Table I.

TABLE I. THE PARAMETERS OF GEO SYSTEM

GEO Satellite	
Parameters	Value
Longitude	116°
Altitude	35786km
Carrier Frequency	20GHz

Bandwidth	40kHz
Antenna radiation pattern	ITU-R S.672
Transmit power	-8.5dB(W/MHz)
Min elevation angel	10°
GEO Earth Station	
Parameters	Value
Longitude	116°
Antenna radiation pattern	ITU-R S.1428
Antenna diameter	1m

The parameters of LEO system are shown in Table II.

TABLE II. THE PARAMETERS OF LEO SYSTEM

LEO Satellite	
Parameters	Value
LEO constellation	OneWeb
Carrier Frequency	20GHz
Bandwidth	40kHz
Antenna radiation pattern	ITU-R S.1528
Transmit power	8dB(W/MHz)
Min elevation angel	10°
LEO Earth Station	
Parameters	Value
Latitude and longitude	GEO earth station
Antenna radiation pattern	ITU-R S.1428
Antenna diameter	2.4m

B. Downlink interference from LEO to GEO

We analyzes the influence of the existence or inexistence of exclusion angle on the downlink interference situation, and simulates the change of the EPFD value when the exclusion angle is set to be 8 degrees/10 degrees/12 degrees. The effect of the exclusion angle strategy can be seen in Fig.8.

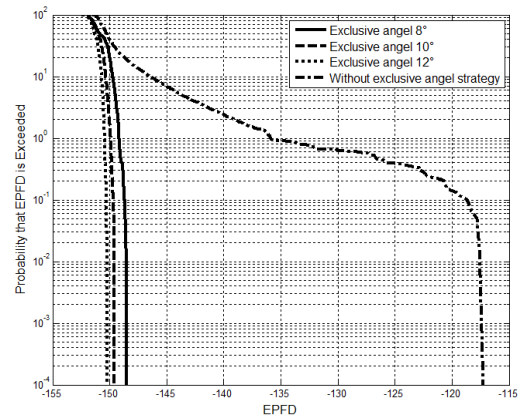


Fig. 8. The interference from LEO to GEO

The exclusive angel strategy makes great decrease of the downlink interference. The bigger the exclusive angel is ,the

less the downlink interference is. It is because that when LEO enters the exclusion zone, LEO does not allow to transmit signal. The strategy may eliminate the situation of in-line interference radically.

However, the coverage of LEO is decreased, we simulate the maximum and minimum latitude of exclusive zone to reflect the range of non-communication zone. It is shown in Fig.9.

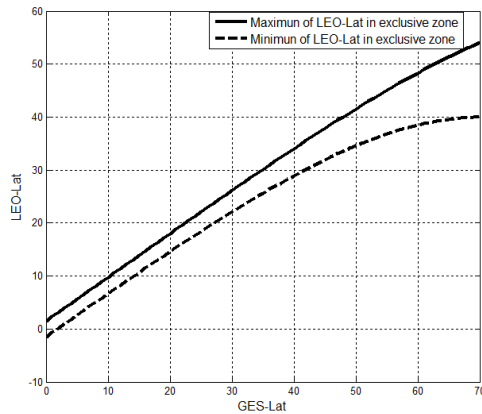


Fig. 9. LEO latitude range in exclusive zone

The non-communication zone is about 3° latitude range when the GEO earth station on the equator, and the degree of latitude range gradually increases, the maximum is 14° when the GEO earth station is set in the polar point. However, there is almost no GEO earth station in the polar area.

C. Downlink interference from GEO to LEO

We simulate the interference from GEO to LEO for 'the worst case'. The result is shown in Fig. 10.

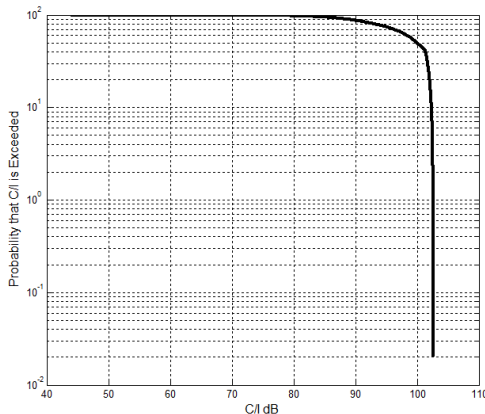


Fig. 10. The interference from GEO to LEO

In the 'worst-case' analysis, we can draw a conclusion that there is little downlink interference from the GEO to LEO. It is mainly because that the altitude of GEO is 36,000 km which is 30 times the altitude of LEO, and the transmission loss is proportional to the square of the transmission distance. In

addition, the antenna pattern and satellite transmit power are also playing an important role for simulation.

CONCLUSION

In this paper, We simulate the downlink interference from LEO system to GEO system with the exclusive angel is 8 degrees/10 degrees/12 degrees. The simulation results show that the exclusive angel strategy is valuable of avoiding the in-line downlink interference, and making a sharp decrease of EPFD. While its impact is limited, and it will bring the non-communication zone. However, it may use satellite routing technology to solve this problem. In addition, we verify that there is little downlink interference from GEO to LEO because of the long distance.

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